

❓ **Cloud Storage:**

GCP's object storage service for storing and retrieving unstructured data like files and media.

Data is stored as objects in buckets, ideal for backup, archives, and big data.

❓ **Persistent Disk:**

Block storage attached to VMs in Compute Engine or Kubernetes Engine.

Supports SSDs and HDDs, suitable for applications needing high performance and durability.

❓ **Cloud Storage for Firebase:**

Designed for mobile apps, enabling secure file uploads/downloads with recovery support.

Handles intermittent network issues efficiently with built-in resilience.

❓ **Cloud Filestore:**

Network-attached file storage that offers shared file systems for VMs and containers.

Ideal for applications requiring high IOPS and customizable storage capacity.

Databases

❓ **Cloud SQL:**

A managed relational database service for MySQL and PostgreSQL with automated backups and patching.

Ideal for users needing traditional SQL databases without managing infrastructure.

❓ **Cloud Bigtable:**

A NoSQL wide-column database for large-scale, low-latency workloads like analytics and IoT.

Supports billions of rows and millions of operations per second.

❓ **Cloud Spanner:**

A globally distributed relational database that combines SQL features with NoSQL scalability.

Offers high availability (99.999% SLA) and strong consistency for enterprise apps.

🔗 **Cloud Datastore:**

A scalable NoSQL document database using key-value pairs, accessed via REST API.

Best for applications needing scalability without strict consistency on reads.

🔗 **Cloud Memorystore:**

A fully managed Redis in-memory caching service for ultra-fast data retrieval. Used to reduce latency by storing frequently accessed data in memory.

🔗 **Cloud Firestore:**

A serverless NoSQL document database for web and mobile apps with real-time sync.

Offers offline support and live updates across devices and backend services.

Networking components

🔗 **Virtual Private Cloud (VPC):**

VPC lets organizations create logically isolated networks in GCP, similar to private data centers.

It enables secure global communication within cloud resources using Google's internal network.

🔗 **Cloud Load Balancing:**

Distributes incoming traffic across multiple servers or regions for high availability and performance.

Supports autoscaling and load balancing for various traffic types like HTTP, HTTPS, TCP, and UDP.

🔗 **Cloud Armor:**

A security service that protects applications from threats like DDoS, XSS, and SQL injection.

It allows custom rules and IP/geolocation-based access control integrated with global load balancing.

🔗 **Cloud CDN:**

Speeds up content delivery by caching static content at global edge locations.
Reduces latency for users worldwide without regional setup complexity.

🔗 **Cloud Interconnect:**

Connects on-premise networks directly or via partners to Google's high-performance network.

Offers private, secure connections using either Dedicated or Partner Interconnect options.

🔗 **Cloud DNS:**

A scalable, high-performance DNS service for translating domain names to IP addresses.

Supports both public and private zones for managing internal and external naming needs.

Billing and accounting

1. Resource Usage and Billing

- Using GCP resources like Virtual Machines (VMs), Cloud Storage, and APIs typically incurs charges.
- Costs vary depending on the resource type, usage duration, region, and pricing tier.
- GCP provides the **Billing API** and tools to manage, track, and control these charges efficiently.

2. Billing Accounts

- A **billing account** stores payment information and is required for using most paid GCP services.
 - Each **GCP project must be linked to a billing account**, unless it only uses free-tier services.
 - A billing account can be associated with one or more GCP projects for centralized cost management.
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3. Types of Billing Accounts

- There are two main billing account types in GCP:

a) Self-Serve Account

- Paid using a **credit card or direct debit from a bank account**.
- Charges are **automatically** deducted based on resource usage.
- Commonly used by **individual users, startups, and small businesses**.

b) Invoiced Account

- Bills are **sent to the customer**, usually **monthly**.
 - Payment is made manually or through the customer's finance department.
 - Typically used by **large enterprises** and customers with high or complex usage patterns.
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4. Billing Budgets and Alerts

- GCP allows users to define **budgets** for their projects and set **alerts** when spending exceeds thresholds.
 - Helps in **monitoring costs** and avoiding unexpected charges.
 - You can configure budgets via the **Billing > Budgets & alerts** section in the console.
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5. Budget Features

- Set a **monthly, quarterly, or custom spending limit**.
 - Define **email alerts or programmatic alerts** using Pub/Sub when costs exceed a percentage of the budget.
 - Helps enforce **financial discipline** and supports **cost forecasting and planning**.
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6. Integration with Resource Hierarchy

- Billing accounts can mirror GCP's resource hierarchy (organization > folders > projects).
- This allows organizations to **track spending per department, team, or product line.**